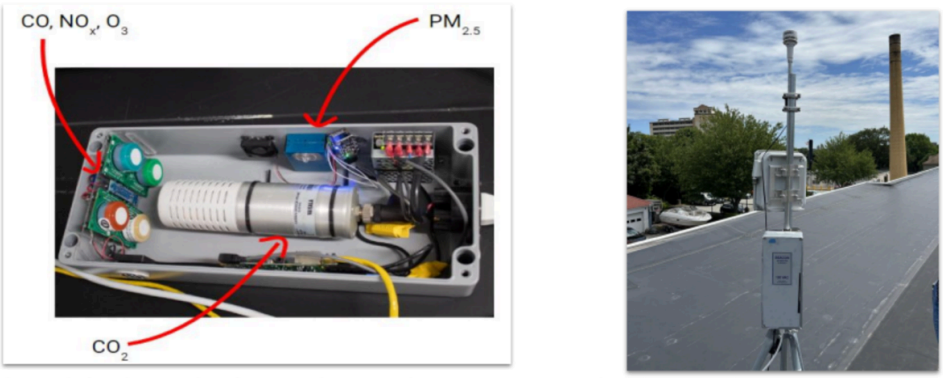
II. Breathe Providence NO_2 Calibration

Overview

Calibration of the Breathe Providence network is a vital process that ensures the accuracy and precision of gas measurements across the network. Our 25 $BEACO_2N$ nodes spread throughout the city of Providence can measure carbon dioxide (CO_2), particulate matter ($PM_{2.5}$), ozone (O_3), nitrogen dioxide (NO_2), nitric oxide (NO), and carbon monoxide (CO). Each is also equipped with a humidity sensor and a temperature sensor within the sensor box itself.



Above, left: The inside of a Breathe Providence $BEACO_2N$ node (Photo credit: Grace Berg).
Above, right: A Breathe Providence collocated $BEACO_2N$ and QuantAQ Node in Providence

Figure 1: See above captions.

Minutely sensor data can be accessed through the $BEACO_2N$ web portal at this link. The back end of the $BEACO_2N$ web portal provides calibration for select gasses, including carbon dioxide (co2_corrected_t_drift_applied) and carbon monoxide (co_corrected). All other gasses are left for us to calibrate ourselves using our own methods.

Ozone, nitric oxide, carbon monoxide, and nitrogen dioxide are measured via Alphasense electrochemical sensors at the top of the sensor box (see figure 1). For these sensors (outside of carbon monoxide, which as mentioned has a calibration in the backend), we look at the wrk_aux signals as the raw variables used to calculate gas measurements. These variables are measured in volts (V) and capture the difference between readings from the working electrode, which is exposed to ambient air and reacts to the target pollutant, and the auxiliary electrode, which is not exposed to the air and captures background/environmental effects, often responding to humidity and temperature. NO_2 specifically is measured via an Alphasense NO2-B43F electrochemical sensor.

Due to sensor drift and manufacturing variability for each Alphasense sensor, it is required that each node utilizes a unique calibration scheme that can translate the raw voltages into accurate gas measurements.

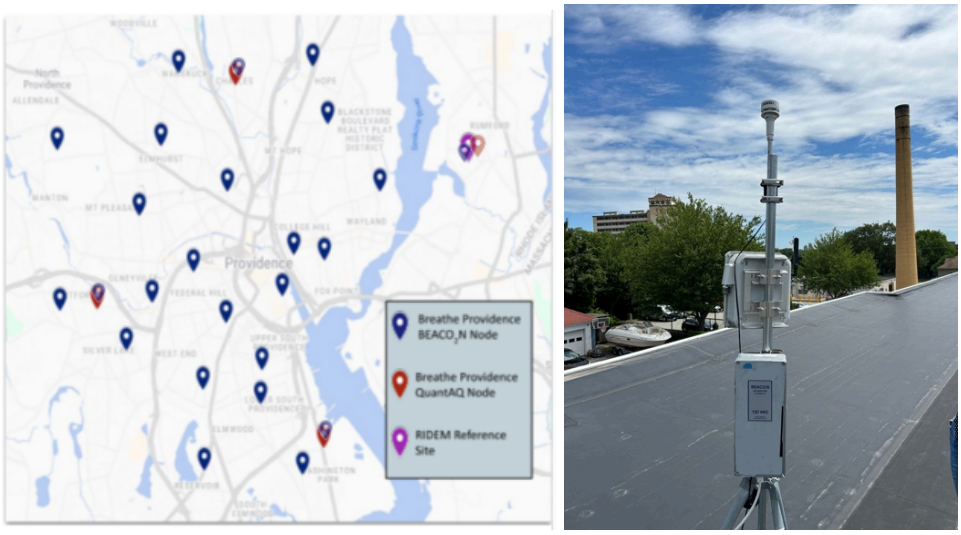


Figure 2: Left: Map of nodes in the Breathe Providence network. All four QuantAQ ("Quant") sites are collocated with BEACO$_2$N sensors. Additionally, the eastern most site at Myron J. Francis contains a BEACO$_2$N sensor, Quant sensor, and a Rhode Island Department of Environmental Management (DEM) sensor. Right: Collocated sensor site at Providence Housing Authority in western Providence, containing both a BEACO$_2$N and Quant sensor.

Calibration Process

In situ low-cost air sensor calibration requires gas measurements from a regulatory grade monitoring station to provide the measurements which we calibrate our sensors against. The calibration process can be broken down into two steps:

- *Precision Correction:* Ensures that all sensors are either in the same background air or calibrate each sensor to a common background.
- *Accuracy Correction:* Calibrating all sensors to a reference instrument after precision correction occurs.

For the Breathe Providence network, all 25 BEACO$_2$N sensors were collocated in the summer of 2022 at a Rhode Island Department of Environmental Management (DEM) site in East Providence. The site is located at Myron J. Francis Elementary School (MJF) and is the only DEM site in the Providence area that can measure NO_2. This collocation took place from July - October 2022.

While this was generally effective to understand sensor offsets for various gas measurements, no seasonal variability was captured during this collocated period which could impact calibration efforts, particularly in colder months where NO_2 concentrations are higher. Generally, collocation is an expensive effort in terms of labor and cost, as all sensors must be physically moved across the city while also ensuring there are enough resources (i.e., power

outlets, physical space at the site) to properly collocate. Data collection is also sacrificed if a sensor must be removed from its designated site.

A remote calibration scheme (one that allows sensors to remain in the field for all parts of the calibration process) at this point is ideal for the Breathe Providence network as all sensors have been deployed. My calibration strategy that I will describe is based on work by Winter et al. 2025, which I highly recommend any reader of this also review.

Precision Step and the QuantAQs

The Winter et al. precision correction relies on the presence of a network of reference grade air sensors. Using these sensors, time periods of well-mixed air are identified and are used for training data for the calibration. We assume that if a (well dispersed) network of reference grade sensors output similar measurements over a given time period, the air is homogenous across the entire city. This allows us to calibrate sensors that are not immediately collocated with a reference instrument during those time periods.

As mentioned previously, Providence contains only one DEM site that can measure NO_2 for the entire city and is located in East Providence. This singular reference site prevents the Winter at al. approach from being used directly, and instead we must rely on a different type of sensor for the identification of well-mixed time periods.

Alongside our 25 BEACO$_2$N sensors, Breathe Providence maintains four QuantAQ Modulair sensors ("Quants") that are collocated with various BEACO$_2$N sites. These sensors are strategically located along the outer edges of the city, and can measure NO_2, O_3, NO, CO, PM_1, $PM_{2.5}$, and PM_{10} at a minutely resolution. They utilize the same Alphasense sensors that are used in the BEACO$_2$N nodes but are equipped with a backend laboratory-based calibration allowing final concentration values to be uploaded directly to the QuantAQ portal. These sites are located at:

- Department of Public Works (DPW). This site is located in southern Providence.
- Myron J. Francis Elementary School (MJF). This site is located in the city of East Providence (as opposed to east side of Providence).
- Providence Emergency Management Agency (PEMA). This site is located in northern Providence.
- Providence Housing Authority (PHA). This site is located in western Providence.

Each gas measured by the Quants feature a separate calibration, with the NO_2 calibration being a hybrid of two regressor models. The Quants have a published NO_2 accuracy within 5 parts per billion (ppb) or 35%, whichever is greater. In the past, I personally have had questions about the accuracy of the Quants and their backend calibration schemes. But under the most recent calibration (that is currently applied to all four Quants) we generally trust the measurements that are used. This topic is explored and justified later in the writeup.

Thus, for my NO_2 calibration scheme described here, we use the Quants' hourly NO_2 measurements for the identification of well-mixed time periods. For reasons that are discussed later in the writeup, we only utilize three of the four Quants (DPW, PEMA, and PHA) to identify the well-mixed time periods. This has to do with the fundamentally different distribution of NO_2 seen at MJF which skews the identification of well-mixed time periods.

Accuracy Step and the QuantAQs

For the accuracy correction, where we train a machine learning model to translate $BEACO_2N$ readings into gas concentrations, we calibrate each site to its nearest Quant. We exclude data from the MJF DEM site for the same reason that we exclude MJF Quant from the identification of well-mixed time periods. The MJF DEM and MJF Quant site are both skewed downward relative to sites more central to the city. Therefore, we would see skewed measurements in the final calibration. This is justified later in the writeup as well.

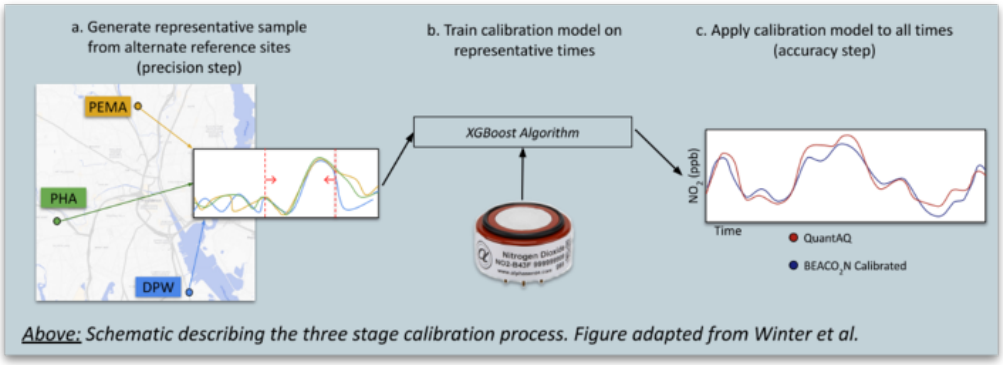


Figure 3: Schematic describing the calibration process pulled from my 2025 Summer UTRA Poster.

GitHub Repository

A calibration script was created that encompasses the entire calibration data pipeline. The link for the repository can be found here. The script downloads all necessary Quant and $BEACO_2N$ data using their respective APIs and applies both the precision and accuracy correction steps. It outputs a final CSV of calibrated data at all 25 $BEACO_2N$ sites and a PDF report with various visuals. These visuals describe details on the calibration process, as well as summary statistics for the final dataset.

The README file in the repository gives specific instructions on how to run the code. Of note, a Quant API key is needed to download data from the Breathe Providence Quant network. Additionally, it is recommended that the "END_DATE" parameter is set to the most recent date possible, as to allow maximum data quantity for training. The following sections will describe the general pipeline and specific design choices made.

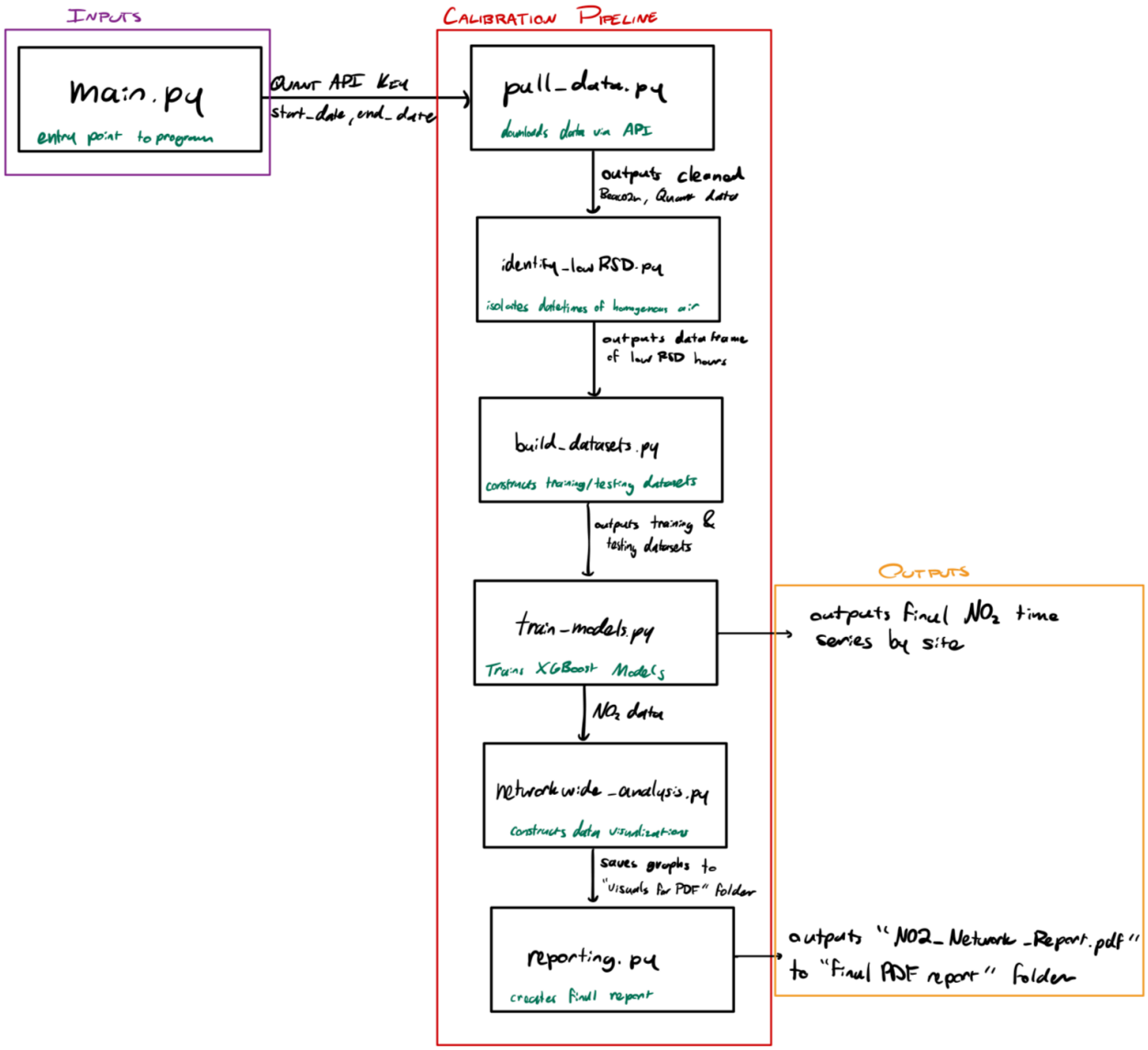


Figure 4: Rough sketch of the calibration pipeline. Outputs are within the yellow box and are outputted at different points of the pipeline.

Main.py

Main.py is where the user enters specific inputs for the script and is designed to be the only file that a user should have to interact with. As mentioned previously, the start and end date must be provided into the program. At the point of writing this, the earliest date that can be entered is October 28, 2025, as this is the first date when all four Quants were continuously online. This may be updated in the future, as will be discussed in a later section as the three "important" Quants (DPW, PEMA, PHA) have been online since late 2024. Error checking is in place to ensure that enough data is downloaded, and the start and end dates are valid.

The start and end date are taken in UTC, where the start date will begin at 00:00:00 UTC (12am), and the end date will be taken at 23:00:00 UTC (11pm). Thus, the end date can be the most recent day where 23:00:00 UTC has occurred. It is (at this point) recommended to keep the start date as early as possible to ensure as much data can be run through the models. These parameters are included primarily to ensure that calibrated data can be downloaded in as close to real time as possible by updating the end date variable.

A Quant API key is also included as a parameter. Only those who are members of the Breathe Providence group on the QuantAQ portal can generate an API key. This can be done by navigating to the "Developer" page on the QuantAQ portal and clicking the blue "Generate a New Key" button.

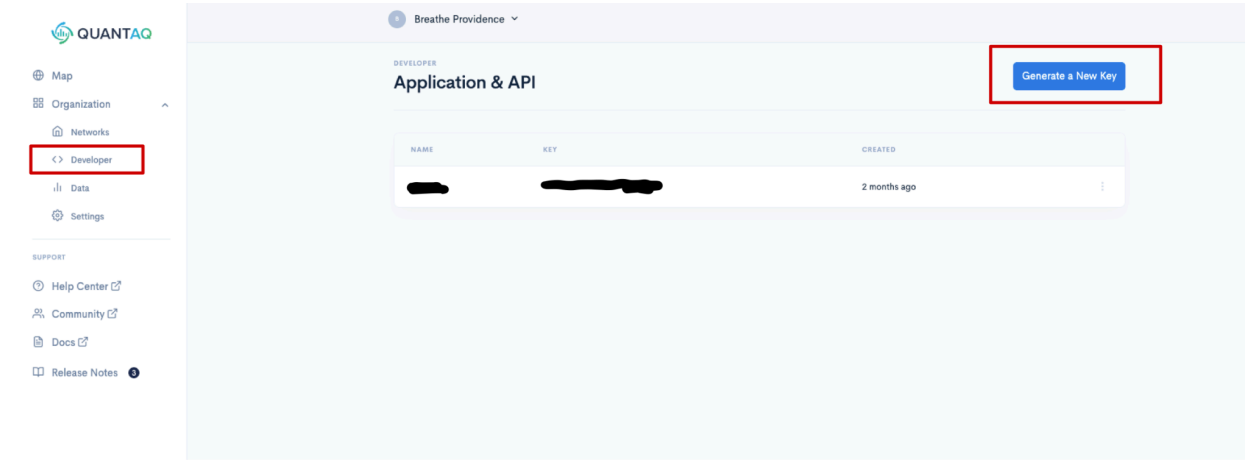


Figure 5: How to generate a Quant API Key from the web portal. If you are a Breathe Providence researcher and need to be added to the group, reach out to Meredith (meredith_hastings@brown.edu) to be added.

While we intend on publishing NO_2 datasets, I recognize this API key can be a bottleneck for accessing data (for those not in the Breathe Providence network). Unfortunately, at this time, there is no way to get around needing this key.

While not recommended, users can also experiment with adjusting which Quant site each $BEACO_2N$ sensor is calibrated to by changing the sensor_to_site dictionary (simply change the short string representing the Quant site).

Users can begin running the script by simply running the main.py python file in their IDE (which is typically done through some sort of play button). Updates will be printed to the terminal as the pipeline files are run.

Pull_data.py

Pull_data.py downloads raw $BEACO_2N$ and Quant data with their respective APIs and cleans the data before moving onto the next step in the pipeline.

The Quant API has an endpoint which automatically averages minutely readings into hourly data but is limited to downloading about 31 days of data in one call. 75% of minutely data in a single hour must be present to return a resampled value per the Quant help center. Thus, the program downloads the data in chunks for each of the four Quant sites, which can be seen in the terminal while running. As of now, historical resampled data for the MJF site ("recalibrated data/mjf_cleaned.csv") is "weaved" into data downloaded from the API as the sensor received the updated NO_2 calibration scheme later than the other Quant sites.

The BEACO$_2$N API also offers an endpoint for data that is resampled hourly. This API is much more straightforward and can download all the data in just one API call per site. Warnings are provided for specific sites where low data coverage is seen.

Identify_low_rsd.py

The identify_low_rsd.py file is responsible for identifying hours of time where the air in Providence is homogenous. In line with Winter et al., we generally aim to construct a time series of relative standard deviation (RSD) values. Thus, we calculate the RSD (standard deviation, scaled by the mean and multiplied by 100 to yield a percentage) spatially between the three sites each hour. We then isolate the bottom 10 percent of the RSD time series to use as our time periods of well-mixed air.

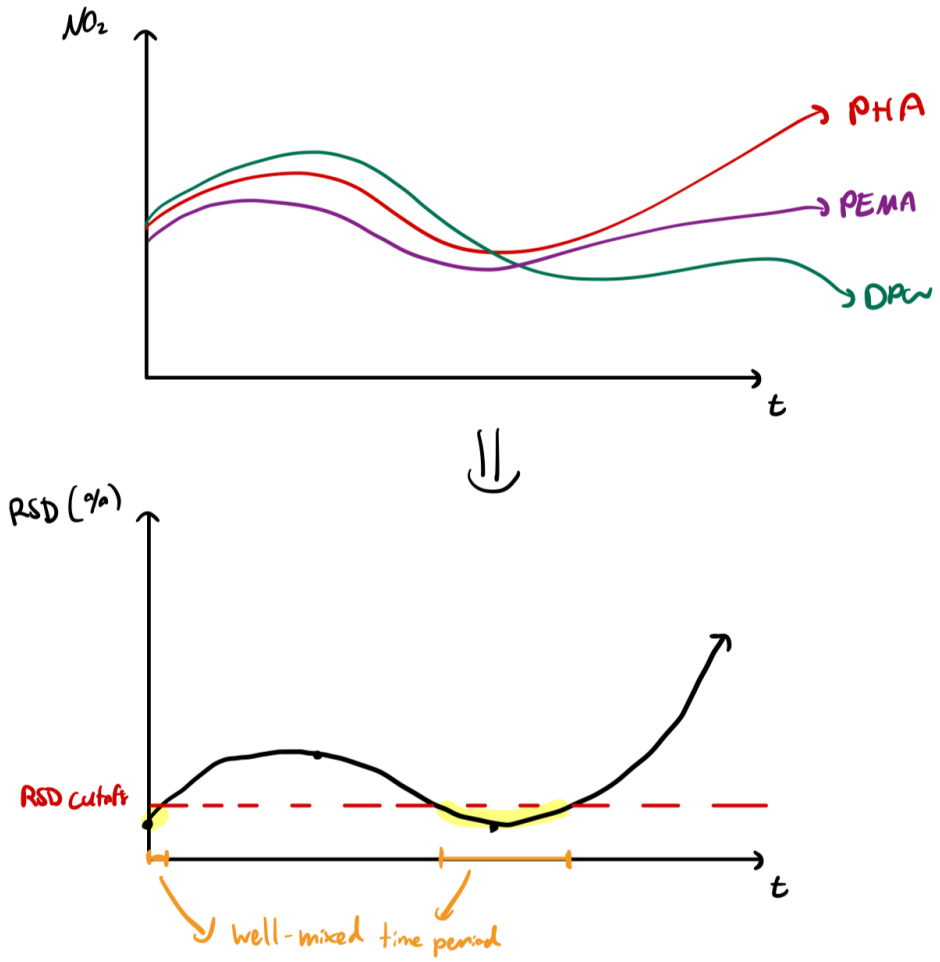


Figure 6: General schematic for identifying time periods of well-mixed air. The RSD cutoff is calculated from the lowest 10% of RSD values over the time series. MJF is excluded from identifying the low RSD values.

This specific approach had an unintended impact on the calibration process. The NO_2 values at the three Quants during these low RSD time periods saw very few values in the 5-15ppb range. This is problematic as these are the NO_2 values that will eventually be used in the training data. If we saw very few 5-15ppb NO_2 values in the training data, the model would not be able to accurately predict NO_2 values in that range for the testing data.

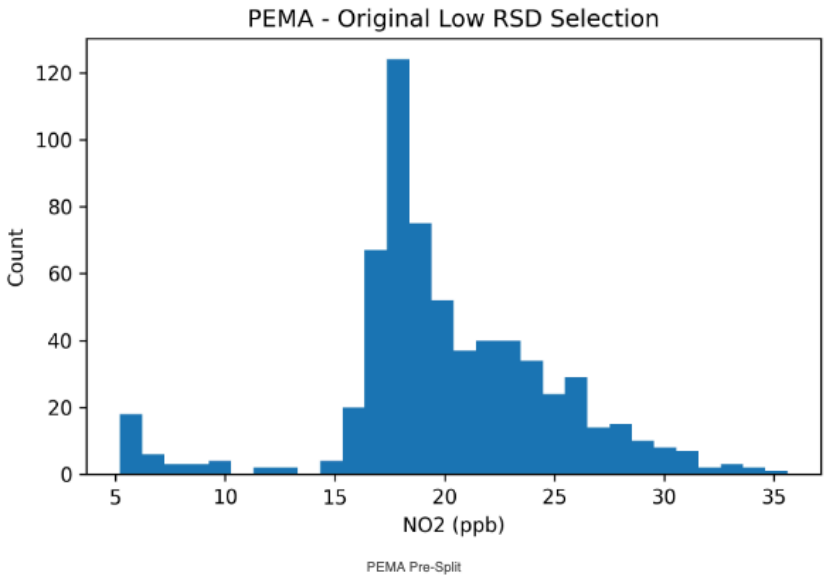


Figure 7: NO_2 distribution at PEMA Quant during the lowest 10th percentile RSD cutoff. Very few NO_2 values are present in the 5-15ppb range. This trend was also present in DPW and PHA and can be seen in the final PDF report generated from the script.

Thus, an altered approach was needed to include these lower NO_2 values in the training data. We take the spatial NO_2 average at the three sites at each hour, and split the data based on whether the average is greater or less than 15ppb. We then take the lowest 10th percentile of RSD values from the <15ppb data, and the lowest 10th percentile of RSD values from the >15ppb data. We then combined the hours identified for each of these two regimes as the final low RSD time periods (which still represents 10% of the total data).

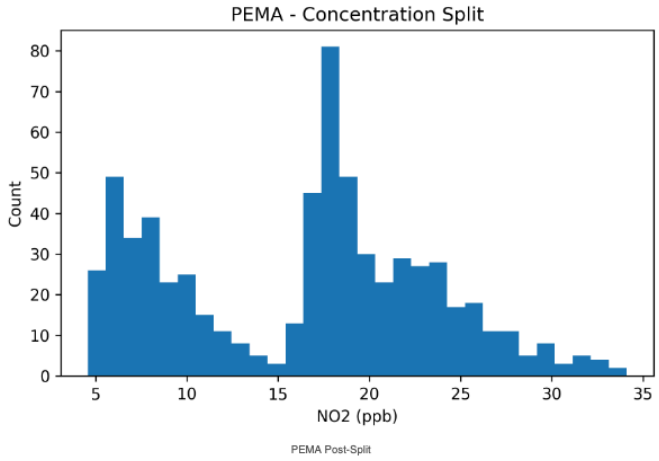


Figure 8: NO_2 distribution at PEMA Quant after using the 15ppb split and lowest 10th percentile cutoff. Using this approach, we see a wider variety of NO_2 values that will be seen in the training data.

The main drawback of this approach is the sacrifice of some spatial homogeneity when forcing the inclusion of values in the lower regime. Based on testing though, this sacrifice is worth making as it dramatically increases testing performance of the machine learning models. The RSD cutoff for both sections remains relatively low. This approach has some interesting effects on the MJF site though, as will be discussed later.

This section of the script will return a dataframe of specific hours where the air is well-mixed that is then used to create training and testing data.

Build_datasets.py

We then proceed to construct the training and testing datasets for each BEACO$_2$N node. Each BEACO$_2$N sensor is calibrated to its nearest Quant site. Notably, only two BEACO$_2$N sites are calibrated to the MJF Quant, those sites being the MJF BEACO$_2$N, and Blackstone Residence on the east side of Providence.

From the BEACO$_2$N data, we include the features no2_wrk_aux, co_wrk_aux, no_wrk_aux, o3_wrk_aux, relative humidity, and temperature as predictors. We include working-auxiliary variables of other gasses (NO, O_3, CO) as they can act as cross-pollutants that impact sensor output for NO_2. This is the approach Winter et al. took for their model, although they did not include carbon monoxide. There is additional discussion at the end as to whether future versions of the script may change parameters that are included in the model.

The testing data takes the BEACO$_2$N parameters (listed above) and combines them with their nearest Quant's NO_2 value as the final "true" value. The testing data for each site contains the entire time series of data, and the training data is simply the testing data but filtered down to only include data during the identified low RSD time periods. Hence the training data is present in the testing data, in line with the Winter et al. paper. This is because we wish to ensure accuracy of the entire time series of data, as well as the fact that the training data only makes up 10% of the time series and does not largely skew the final performance statistics.

Train_models.py

Here we train our models to predict NO_2 values. As stated before, our features are no2_wrk_aux, co_wrk_aux, no_wrk_aux, o3_wrk_aux, temp, and rh, and we aim to predict NO_2 of the nearest Quant during the low RSD time periods. The model that is learned is then applied across the entire time series.

Past calibration schemes, including those done by Winter et al. and former Breathe Providence calibrations, would use multiple linear regression (MLR) to predict concentration values from the working-auxiliary inputs. These models are beneficial because of their simplicity, non-black box nature, and likely better extrapolation to out-of-distribution events. For various reasons that will be covered later in the writeup, MLR is not a great model of choice here due to non-linear behavior of the Alphasense sensors.

Thus, we must utilize a more sophisticated machine learning model that can deal with nonlinear interactions between features. XGBoost (extreme gradient boosting) is the model of choice that with past testing seems to perform best with handling calibration. At a conceptual level, gradient boosting is a tree-based machine learning model that iteratively builds more trees to make better predictions. The python XGboost package is simple to use and is used in the calibration code.

With such an advanced model relative to a simple MLR, it is important that hyperparameters are properly tuned to ensure overfitting does not occur. I created a notebook that could hold out a validation month while testing combinations of hyperparameters on our XGBoost model. The GitHub link for this project is here, and although the code is a bit raw it can certainly be run by a user with different combinations of hyperparameters.

The user can simply change the param_grid variable to test different hyperparameter ranges on the training and testing data. It should be noted that each new hyperparameter value will increase the time it takes to run the notebook by several times. Here, the training and testing data is compiled from Oct 2025 through July 2026, and thus covers most seasonality seen in the Providence area.

The validation_month variable is the month that is held out during the testing and training process, and that the model is applied to. We want to see ideal performance on this fully unseen month to confirm that we are not overfitting to data seen during training. Users simply run the notebook with their validation month and hyperparameter grid specified. The combinations produced will be saved to the "validation summary.csv" file in the outputs folder.

	combo_idx	max_depth	learning_rate	n_estimators	subsample	mean_r2	mean_rmse	mean_val_r2	mean_val_rmse
23	23	4	0.05	150	1.0	0.6262407291176767	3.4524104143030527	0.1412078779484329	3.1511473331579105
24	24	4	0.05	200	1.0	0.6262256664459668	3.4512154569544706	0.14114120570199554	3.1582368535405188
22	22	4	0.05	100	1.0	0.625132646495577	3.461102677431901	0.13749868117925315	3.1447724654926263
28	28	4	0.1	150	1.0	0.6221249494521501	3.468896707741871	0.11184739778249375	3.2072442656418314
27	27	4	0.1	100	1.0	0.6218770715735679	3.470919518570522	0.110508427246572	3.2060123839365913
29	29	4	0.1	200	1.0	0.6218238872440917	3.4699919096442344	0.11050123579612792	3.210641340177372
26	26	4	0.1	75	1.0	0.6212727972313852	3.4748367808071676	0.1093818856134543	3.2072076077086986

Figure 9: Validation summary table. In the table produced in the code, the values are sorted by the mean root mean square error (RMSE) seen at all collocated sites. So here, we are only considering DPW, MJF, PEMA, and PHA. Each combination of hyperparameters receives a unique combination index.

"compiled data.csv" in the main repository compiles several validation summary tables into one CSV file. The full parameter grid was tested on all eight complete months available in the data at the time (November 2025 - June 2026) and combined into one CSV. These values are then averaged for each combination index in the final cell of the notebook. This final cell of the notebook determines which hyperparameter combinations, on average, produce the best performance on validation months. Thus, the best combinations here will describe strong models that are better extrapolating temporally.

Based on this table, it was decided that the ideal hyperparameter combination here is:

- n_estimators = 150
- max_depth = 2
- learning_rate = 0.1
- subsample = 1.0
- colsample_bytree = 0.8

As of writing, this is the combination that is in the main calibration repository and until further notice will continue to be used. There is a non-zero chance that this is changed in the future, but it will simply be reflected in the GitHub repository. If anyone else takes on this project, I encourage more experimentation with the hyperparameters.

Networkwide_analysis.py and reporting.py

These are the two python scripts that are responsible for the creation of the final PDF that is outputted alongside the calibrated NO_2 time series. Networkwide_analysis.py generates various data visuals based on the calibrated dataset itself. It primarily utilizes the matplotlib package in python, but also for one visual uses the GeoPandas package. All visuals generated in the networkwide_analysis.py file are saved to the "final PDF report/visuals for PDF" folder in the main repository.

Reporting.py then compiles visuals from this folder (which additionally stores some additional visuals from the train_models.py), as well as a few tables into a PDF report. For this task, the ReportLab python package is used. The following summarizes all visuals that are saved to this report (at the time of writing):

- Quant summary statistics for NO_2 at all Quant sites (DPW, MJF, PEMA, PHA) during given period.
- Quant boxplots for NO_2 at all Quant sites during given period.
- Train test split data amounts by sensor at all 25 $BEACO_2N$ sites.
- Histograms of NO_2 training data at all Quant sites before and after enforcing the 15ppb data split (see figures 7 and 8).
- Training and testing data scatter plots at all four $BEACO_2N$ sites that are collocated with Quant sites.
- XGBoost model performance metrics table (at all $BEACO_2N$ sites).
    - Please see results section on how to interpret this table.
- Network wide summary statistics for NO_2 at all 25 $BEACO_2N$ sites.
- NO_2 boxplot by site, including and excluding MJF and Blackstone Residence sites.
- Median NO_2 by site bar graph, including and excluding MJF and Blackstone Residence sites.

- 95th percentile exposure by site bar graph.
- Median concentration by site on map, scaled medians via z-score.
- Site to site correlation heatmap.
- NO_2 diurnal cycle heatmap.
- NO_2 monthly trend heatmap.
- Hierarchical clustering diagrams between sites.

Results

sensor	site	site_name	train_r2	train_mae	train_rmse	test_r2	test_mae	test_rmse	top1_feature	top1_importance	top2_feature	top2_importance	n_train	n_test
Sensor13	pha	PHA	0.981	0.69	0.95	0.893	1.59	2.27	temp	0.5	no2_wrk_aux	0.289999991655349	528	5467
Sensor12	pha	United Way	0.98	0.73	1.06	0.888	1.65	2.4	temp	0.519999980926513	no2_wrk_aux	0.349999994039535	689	6186
Sensor04	pha	Carnevale	0.975	0.84	1.2	0.877	1.79	2.52	temp	0.469999998807907	no2_wrk_aux	0.370000004768371	689	6164
Sensor10	pha	Zuccolo	0.975	0.85	1.18	0.856	1.91	2.77	temp	0.519999980926513	no2_wrk_aux	0.239999994635581	638	5347
Sensor02	pha	Silver Lake	0.977	0.84	1.14	0.856	1.91	2.72	temp	0.519999980926513	no2_wrk_aux	0.349999994039535	689	6188
Sensor01	mjf	MJF Elementary	0.953	0.94	1.33	0.85	1.44	1.99	temp	0.370000004768371	no2_wrk_aux	0.370000004768371	673	6149
Sensor22	pha	Mt Pleasant	0.972	0.84	1.24	0.849	1.95	2.74	temp	0.600000023841857	no2_wrk_aux	0.300000011920928	646	5133
Sensor17	mjf	Blackstone	0.942	1.02	1.48	0.821	1.54	2.18	no2_wrk_aux	0.430000007152557	temp	0.259999990463256	689	6179
Sensor25	pema	PEMA	0.979	0.74	1.07	0.795	2.0	2.8	temp	0.550000001192092	no2_wrk_aux	0.280000001192092	688	6185
Sensor06	pema	Rochambeau	0.962	0.98	1.46	0.734	2.36	3.2	temp	0.540000021457672	no2_wrk_aux	0.270000010728836	589	5047
Sensor15	pema	Main St MA	0.968	0.94	1.33	0.731	2.42	3.21	temp	0.519999980926513	no2_wrk_aux	0.319999992847442	602	5640
Sensor24	pema	Providence Coll	0.965	0.95	1.38	0.725	2.42	3.24	temp	0.5	no2_wrk_aux	0.310000002384185	678	6148
Sensor23	pha	RIC	0.954	1.12	1.61	0.693	2.79	3.96	temp	0.509999990463256	rh	0.230000004172325	444	4711
Sensor05	pema	E-Cubed	0.963	0.95	1.42	0.693	2.55	3.42	temp	0.55000001192092	no2_wrk_aux	0.319999992847442	689	6179
Sensor21	dpw	Child & Family	0.974	0.84	1.18	0.624	2.6	3.79	temp	0.509999990463256	no2_wrk_aux	0.379999995231628	689	6188
Sensor18	dpw	Brown-Fox Pt	0.96	1.0	1.46	0.612	2.71	3.85	temp	0.529999971389770	no2_wrk_aux	0.319999992847442	689	6182
Sensor07	pema	Smith Hill	0.934	1.16	1.6	0.597	2.97	3.85	temp	0.469999998807907	no2_wrk_aux	0.280000001192092	500	5277
Sensor03	dpw	Reservoir Ave	0.974	0.84	1.18	0.589	2.72	3.96	temp	0.519999980926513	no2_wrk_aux	0.349999994039535	689	6183
Sensor14	dpw	CCRI	0.966	0.94	1.34	0.584	2.73	3.98	temp	0.569999992847442	no2_wrk_aux	0.289999991655349	687	6186
Sensor20	pema	Rockefeller	0.962	1.0	1.44	0.574	2.97	4.03	temp	0.620000004768371	no2_wrk_aux	0.270000010728836	689	6185
Sensor09	dpw	DPW	0.95	1.1	1.62	0.547	2.93	4.16	temp	0.5	o3_wrk_aux	0.25	688	6186
Sensor16	dpw	South Prov	0.964	1.02	1.39	0.529	2.94	4.24	temp	0.55000001192092	no2_wrk_aux	0.319999992847442	689	6189
Sensor08	dpw	Alpert Med	0.951	1.11	1.58	0.494	2.89	4.31	temp	0.519999980926513	no2_wrk_aux	0.319999992847442	623	5941
Sensor19	pha	Rock Spot	0.879	1.77	2.56	0.427	3.93	5.37	temp	0.469999998807907	o3_wrk_aux	0.189999997615814	529	5231

Figure 10: Calibration results at all sites. In terms of model performance, we only care about the four BEACO$_2$N sites that are collocated with Quants, those being DPW, MJF, PEMA, and PHA.

General model results are summarized in figure 10, pulled directly from the NO_2 network report that is provided when running the calibration script. This particular calibration call was run from dates 10/28/2025 - 8/20/2026. The model generally performs well on testing data relative to past attempts to calibrate and is in line with the Winter et al. group's work at some sites. PHA, MJF, and PEMA all see coefficients of determination (R^2) values of around or above 0.8. Interestingly, MJF sees the lowest test root mean square error (RMSE) and mean absolute error (MAE) of any of the 25 sites. DPW performs quite poorly, but the importance values of DPW indicate that the features with highest importance are temperature and O_3, rather than temperature and NO_2. A later section will discuss these importance values but the dependence on O_3 indicates a hardware malfunction at that particular site. This is supported by inspecting voltage outputs from the Alphasense O_3 sensor relative to other sites (figure 11).

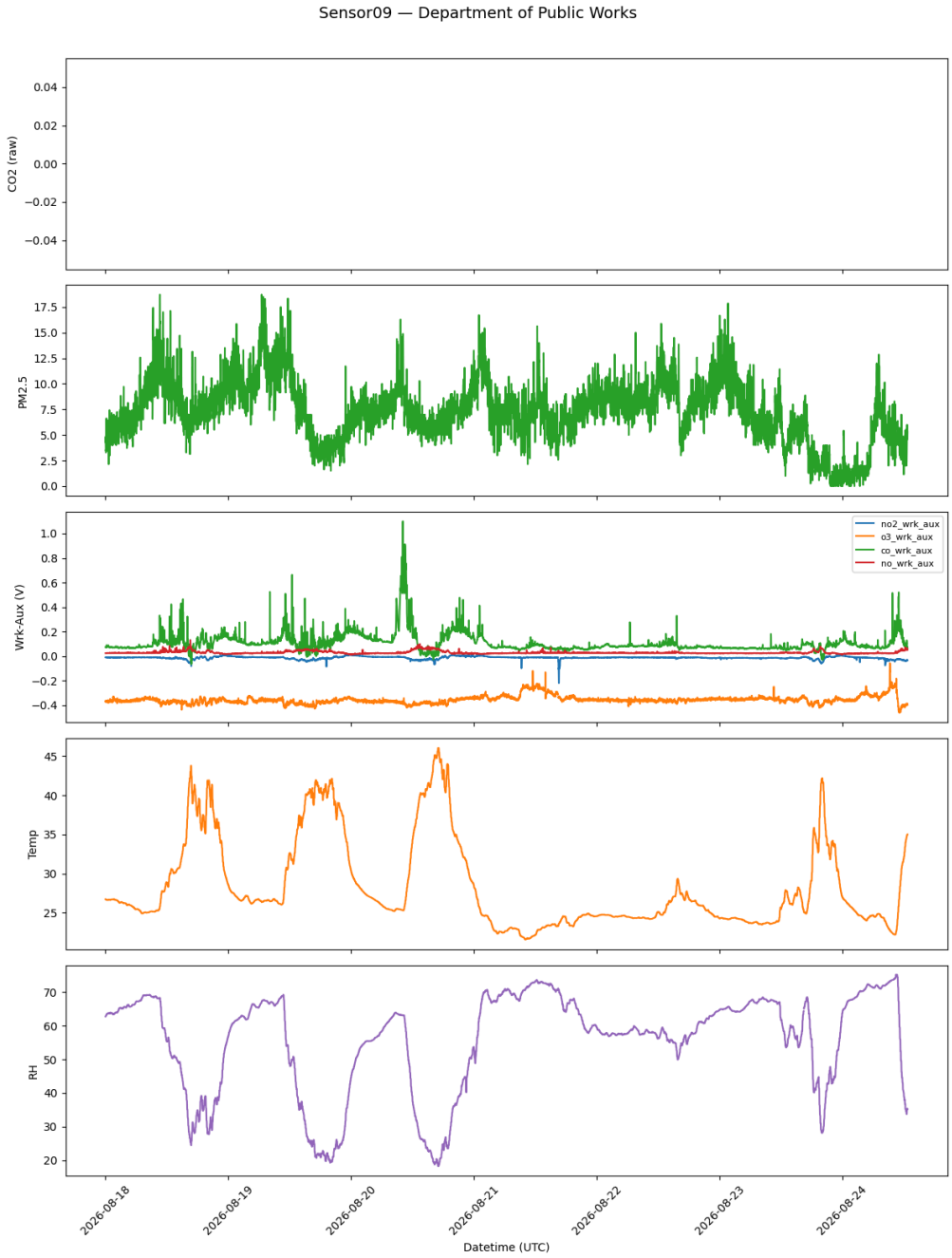


Figure 11: DPW raw sensor outputs. Focus on the middle graph which lays out Alphasense sensor voltage output over the span of a week. Notice the orange (ozone) signal sees strong negative voltage values as low as -0.4. Values this large in magnitude are not seen at any other BEACO$_2$N site for any other pollutant. See the site status script section that describes the code where this visual comes from.

While performance at collocated sites is generally good, some questions arise about the spatial gradient that arises under this calibration scheme. Specifically, the two BEACO$_2$N sites that are calibrated to the MJF Quant site see much lower concentrations relative to the rest of the network by a significant amount.

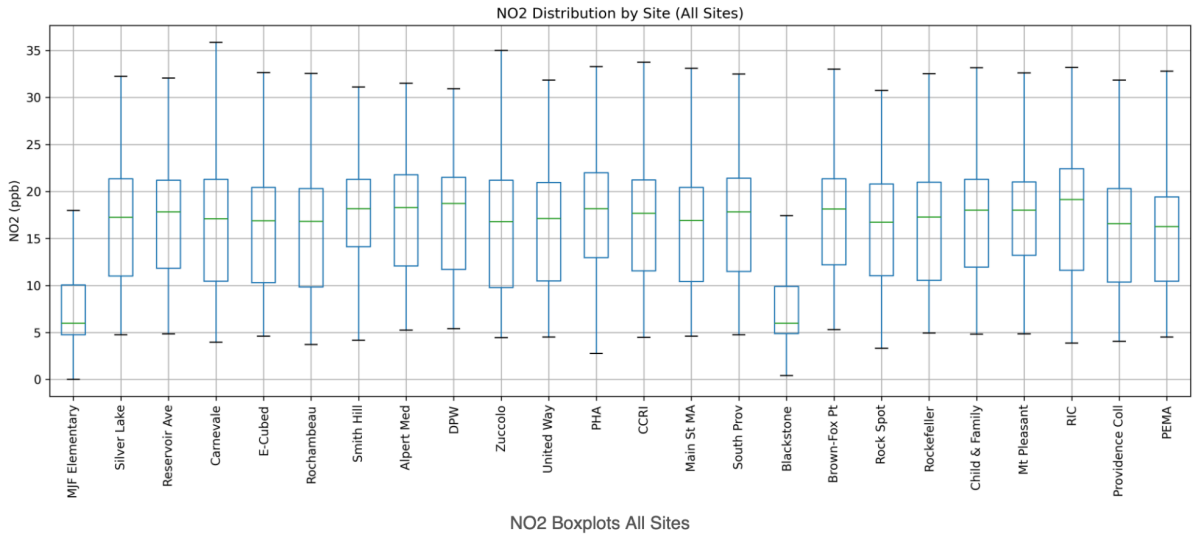


Figure 12: NO_2 boxplot by site. This visual is produced in the PDF report for the script, and here this data is pulled from the same ranges of dates as figure 10. The two sites (The MJF $BEACO_2N$ and Blackstone Residence) that are calibrated to the MJF Quant are clearly visible and obtain a much lower distribution.

Generally I believe the calibrations to be trustworthy due to their collocation performance on the testing data. However, I think this apparent gradient between concentrations in the city versus those sites further east of the city is concerning. The site at Brown-Fox Point (BFP) Early Childhood Learning Center on the East Side of Providence is slightly closer to the DPW Quant than the MJF Quant. When switching calibration of that site to MJF Quant, the BFP site completely shifts its distribution to reflect that of the MJF site.

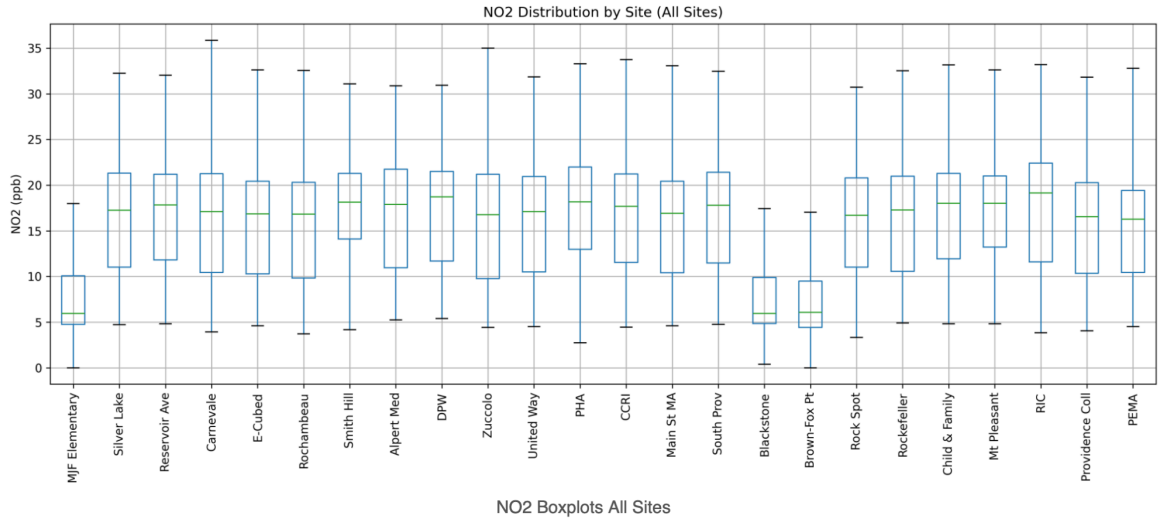


Figure 13: Boxplots for calibrated NO_2 data at each site. Notice now that BFP aligns with other sites that are calibrated to the MJF Quant.

The test R^2 value for BFP increased when switching the calibration to the MJF site which may be interpreted as the model having an "easier" time translating the inputs into the final calibration, but I am not confident that means MJF is the ideal site to calibrate.

The data split being enforced on the MJF site also means that lower concentration values are overrepresented in the training data for the model. This may end up being something that changes in the final version of the calibration, where we would somehow remove this data split from the MJF alone, but not the other sites.

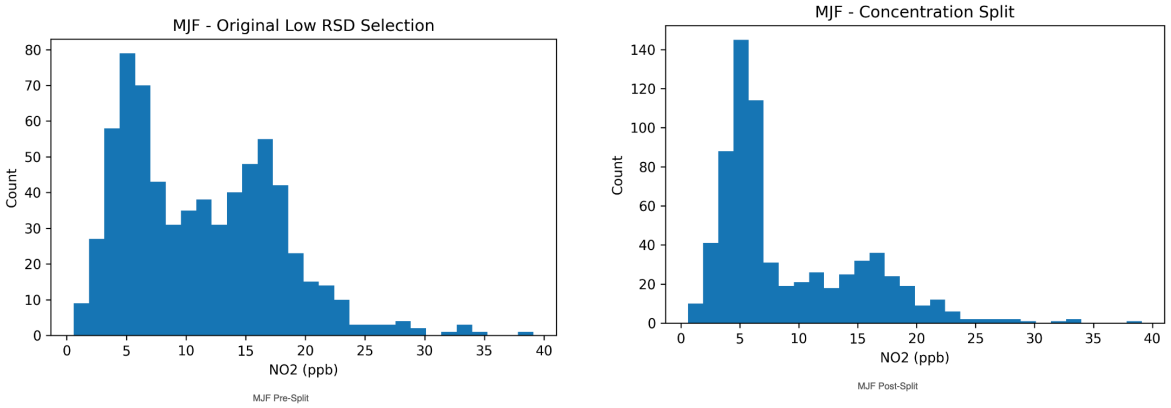
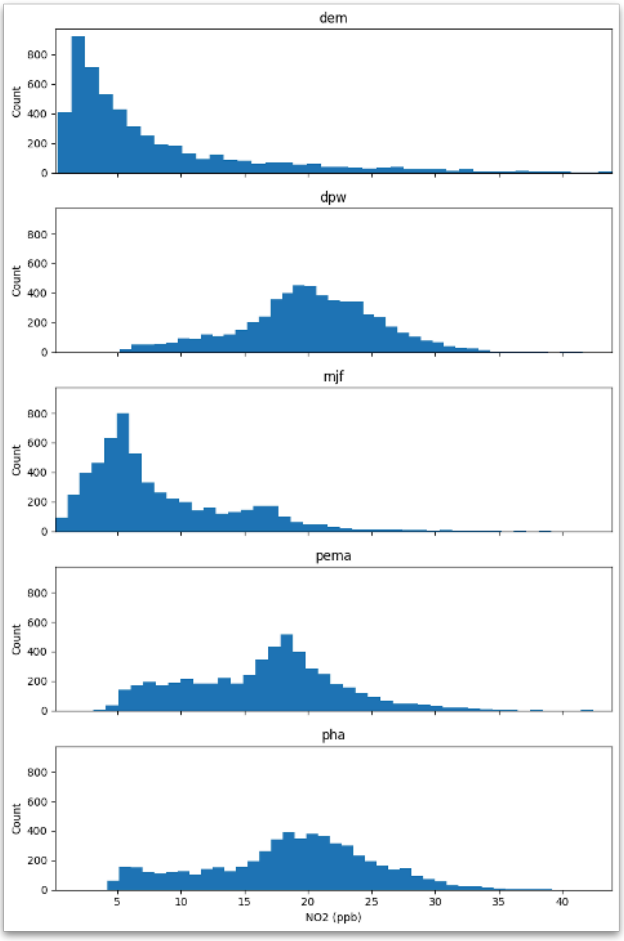


Figure 14: The impact of enforcing the data split on the MJF Quant. The best way I can think of avoiding this issue is dropping the low RSD requirement all together and calibrating the MJF sites separately. A simple random train-test split could work for those sites.

For now, I think this gradient of concentrations must exist in this calibration and is reflective of a real concentration gradient across the city, as evidenced by the Quant distributions themselves.

Why do we not calibrate to the RIDEM site directly?

An ideal calibration would, to some extent, include data from the MJF DEM site, whether that be for the identification of well-mixed time periods (precision correction) or as the true reference to calibrate against (accuracy correction). Neither of these are ideal here as the distribution of NO_2 at the MJF DEM site in East Providence is fundamentally different then the distribution of NO_2 seen more central to the city of Providence.



NO2 Summary Statistics (ppb)

	Count	Missing	Mean	Std	Min	25%	Median	75%	Max	IQR
dem	5184	434	7.66	7.81	0.3	2.4	4.7	9.72	43.9	7.32
dpw	5613	5	20.14	5.42	5.2	17.2	20.2	23.70	41.6	6.50
mjf	5616	2	8.20	5.66	0.1	4.3	6.2	11.10	39.1	6.80
pema	5202	416	16.48	5.97	3.1	12.0	17.2	20.10	42.4	8.10
pha	5616	2	18.56	6.52	4.2	14.6	19.2	22.90	41.0	8.30

Figure 15: NO_2 distributions and summary statistics from Fall 2025 - Spring 2026 among Quants and DEM reference site. Here, DEM and MJF are collocated at the Myron J. Francis site in East Providence.

As seen in Figure 15, the distribution of NO_2 at the MJF DEM site is shifted much lower than those sites seen in the city. In general, the distribution of NO_2 of the MJF Quant and the MJF DEM site are similar and indicate that NO_2 in the city (from sites in the city, DPW, PEMA, and PHA) is much higher than that seen at the MJF DEM site in East Providence.

This is, in my opinion, a serious issue for the city of Providence and the state/city government which uses this MJF DEM site to draw conclusions about NO_2 levels in the city of Providence. RIDEM has generally deemed the NO_2 levels in the city acceptable every five years based on a site that is not located in Providence and within a suburb. RIDEM would claim that this sensor is upwind of the city, and if you can only place one government-grade sensor for a large city that is the ideal strategy to use (rather than a central site which may not be able to record pollution that is further downwind). However, it appears this approach has failed, in my opinion, due to local surface wind patterns not matching the "upwind" claim. RIDEM should consider relocating this site or introducing new sites that measure NO_2 (though this would certainly require a large increase in or higher uptake of funding).

The vastly different distribution of NO_2 at MJF DEM would force the model to produce much lower distributions of NO_2. Even if the Quants were used to identify the well-mixed time periods and then BEACO_2Ns were calibrated to the DEM site, the models would still output what concentration is predicted at DEM (which would then be much lower than the concentrations seen in the city with the Quants).

It would also not be ideal to use DEM in the identification of well-mixed air. This idea is explained in an upcoming section in terms of using the MJF Quant to identify well-mixed air (as the distributions from the DEM and Quant at MJF are in agreement).

Furthermore, using the DEM data would prevent the script from being real-near time. RIDEM often takes months to upload air quality data to the AQS API. Overall, incorporating DEM NO_2 data in the calibration would not be effective and would lead to underestimation of NO_2 concentrations in the city.

Do we trust the Quant measurements?

As previously mentioned, the Quants claim an accuracy of 5ppb or 35%, whichever is greater, and all Quants are currently on the most up-to-date NO_2 calibration scheme. Previous Quant backend calibration schemes would output an NO_2 distribution with a distinct bimodal shape. This made prior calibration attempts from $BEACO_2N$ measurements to Quants quite difficult as there were no variable regimes that would define which peak a concentration would fall near (i.e., there was no way to split data on humidity, temperature, time of day, etc. to split apart the bimodal peak). This calibration attempt is detailed in my summer 2025 UTRA poster.

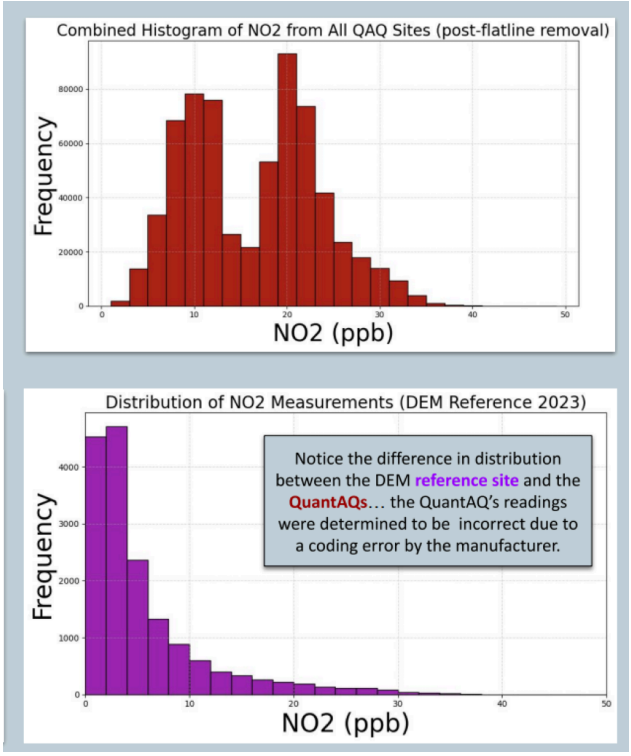


Figure 16: Visual pulled from my 2025 Summer UTRA poster. This was for a prior calibration attempt using the old Quant calibration scheme. Notice the distinct bimodal peak in the distribution of NO_2 data at all Quant sites under the old scheme. I found that it was difficult for any model (XGBoost, RandomForest, MLR) to predict which peak the NO_2 should fall under. QuantAQ had claimed that this was something to do with the k-nearest neighbors components of their old calibration scheme.

This calibration attempt left me skeptical of the accuracy of the Quants and their calibration (even their new calibration). It is worth discussing the accuracy of the Quants outside of what QuantAQ claims.

The best way to analyze the accuracy of the Quants is to compare the readings from the MJF Quant to the MJF DEM site that it is collocated with. While the overall distributions are very similar between the sensors, the R^2 values when comparing NO_2 concentrations recording at

the same time were quite low (less than 0.2). This was initially a disappointing find, but there are reasons to believe that the Quant is accurate enough for our uses.

While the MJF Quant does not perform well relative to the MJF DEM site, the calibrated data from the MJF BEACO$_2$N (when the MJF BEACO$_2$N is calibrated to the MJF Quant) does compare well to the MJF DEM site.

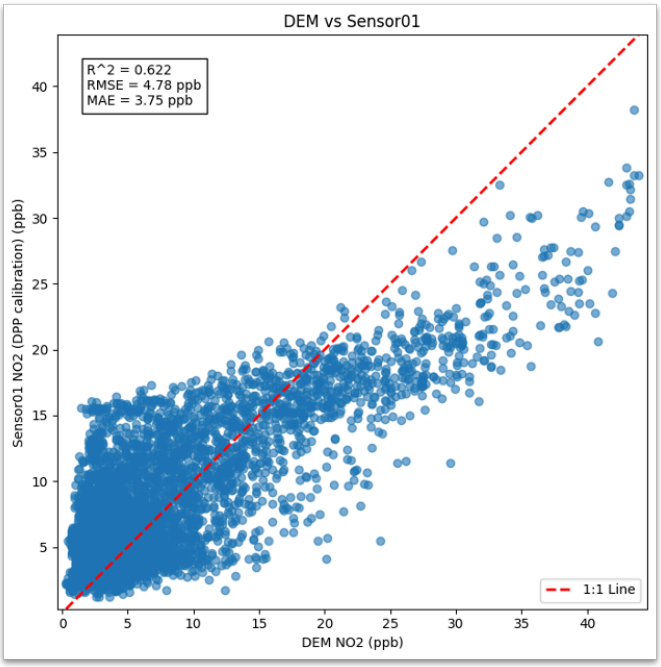


Figure 17: Calibrated BEACO$_2$N data from the MJF site versus the collocated MJF DEM site. The data used here is from October 2025 - May 2026. The coefficient of determination is in an acceptable range here.

Figure 17 depicts the performance of the calibration relative to the MJF DEM site. While error values are generally large, it is likely driven by underprediction at the higher end. While this may indicate calibration may underpredict higher concentration events, we prefer this to overprediction from a policy perspective. Additionally, the R^2 value here is an acceptable range relative to prior works on calibration.

Thus, while the Quants may not have great collocation performance to reference grade equipment, they are still accurate enough to produce a trustworthy calibration for the BEACO$_2$N sites.

Why don't we use the MJF Quant in the identification of well-mixed time periods?

This calibration does not utilize the MJF Quant in the identification of well-mixed time periods, nor do we use the MJF DEM site. An ideal calibration should use all reference grade equipment available to identify these well-mixed time periods to calibrate the whole network, but that approach cannot be used here. This again comes back to the idea that the distribution of NO_2 reflected at MJF through both the Quant and DEM site are much lower than those seen in the rest of the network.

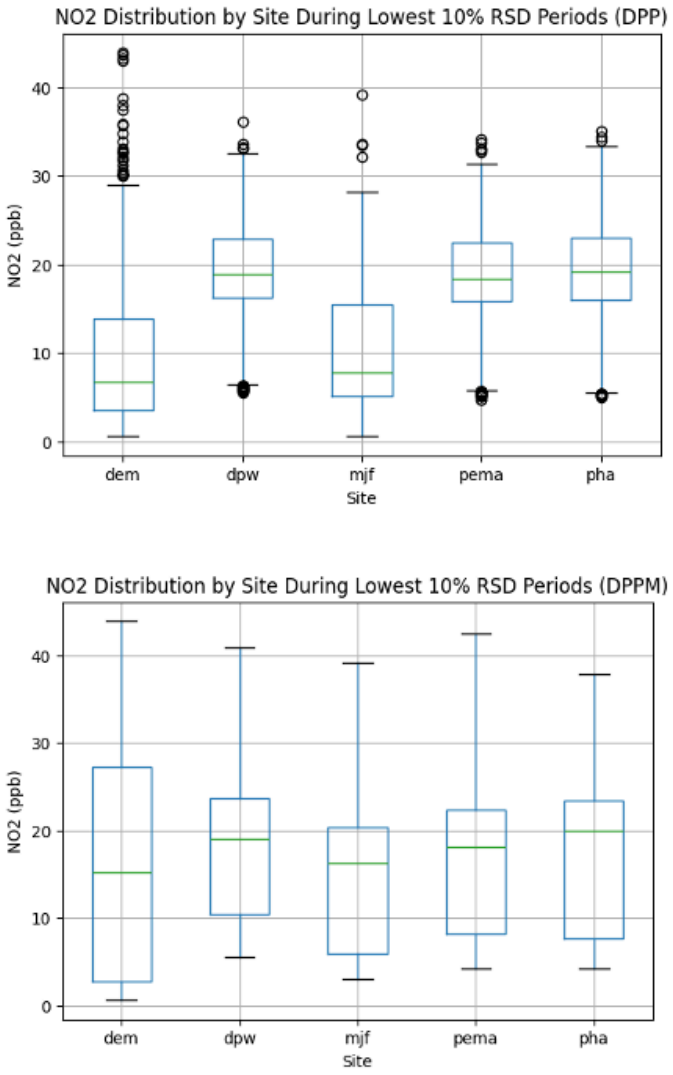


Figure 18: NO_2 distribution by site during lowest 10% RSD time period. This is not using the 15 ppb data split method and acts as more of a general representation. Here, DPP uses only three Quant sites to identify well-mixed air (DPW, PEMA, PHA), and DPPM uses all four Quants (including MJF) to identify well-mixed time periods.

When we include MJF site, the RSD cutoff becomes higher than if it were not included. This indicates that we are sacrificing spatial homogeneity in these time periods to incorporate the MJF site in East Providence. As seen in figure 18, we also see that DPW, PEMA, and PHA see the inclusion of much lower concentration values that do not represent their overall concentrations. These lower concentration values that are seen in the training data will further skew the final NO_2 values downward as well. Here, the bias towards lower values is not beneficial as it skews the entire final distribution lower, not just extreme events.

Why use XGBoost?

The Winter et al. group utilize MLR to calibrate their NO_2 readings. This is a natural choice, as ideally the Alphasense sensors respond linearly to higher concentration values. The downside to this approach is that in higher temperatures, the sensors stop responding linearly.

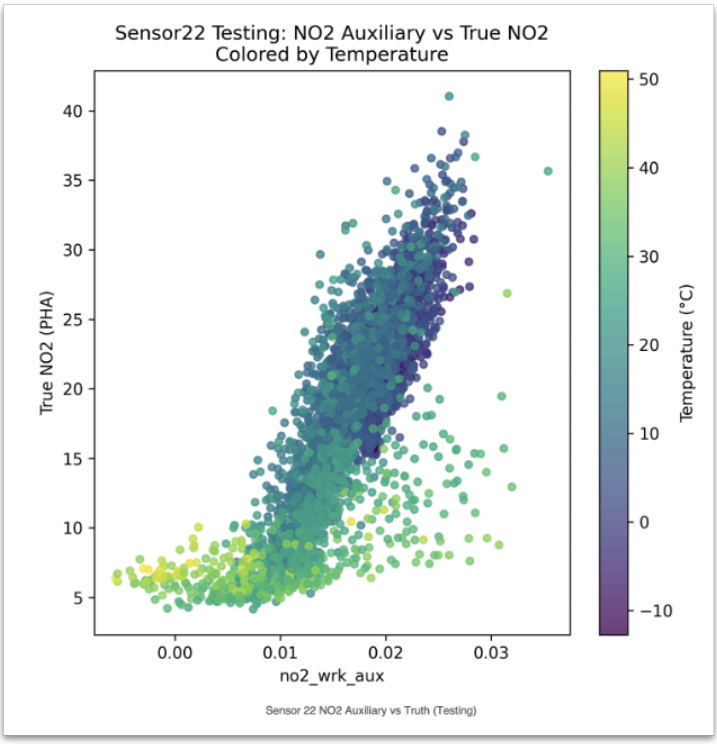


Figure 19: NO_2 readings versus no2_wrk_aux variable at Sensor22 (Mount Pleasant Hardware). Notice the linear shape of the data in the lower temperature ranges, while in higher temperature ranges the data is much more random. This behavior is seen at all sites.

The Winter et al. approach involves removing all data points associated with temperatures above 30 degrees celsius. Breathe Providence cannot take this approach as about 10% of data from October 2025 - June 2026 is above that threshold.

A more sophisticated machine learning model is needed to deal with the non-linear behavior that can occur in higher temperature ranges. Gradient boosting is a natural choice for this and is known for its ability to handle non-linear data. Some experimentation was done with RandomForest, but XGBoost seemed to be the most consistent algorithm. MLR was not effective with the NO_2 data and saw much lower R^2 values at collocated sites.

While XGBoost is effective, it is also a blackbox algorithm that provides little insight on to the actual inner workings of the model. I found it important to explore XAI techniques that ensure the models are behaving as they should (responding to the right BEACO$_2$N inputs), and those findings are described in the next section.

A Note on XAI

Within the PDF report generated by the calibration script, importance values are included in the final table as seen in figure 10. These values are from the XGBoost package and are defined based on gain (essentially how much loss is reduced when splitting on a certain feature). For some experimental versions of the script, I also included permutation importance values but found they were consistent with importance values from the XGBoost package. In the table, the top two highest importance features were included by name and their corresponding importance values.

An interesting finding was that for the vast majority of BEACO$_2$N sites, the parameter with the highest predictive power was temperature. I did not expect this result, as no2_wrk_aux should still be correlated directly with concentration at most temperatures. It is typically the second most important value.

When hyperparameter tuning, I experimented with boosting the weight of the no2_wrk_aux variables and targeting hyperparameter combinations where no2_wrk_aux would take over as the parameter of highest performance. However, these hyperparameter combinations would typically see worse performance.

At the point of writing, we instead target hyperparameter combinations that test well against temporal validation sites. The temperature parameter seeing highest predictive power is most likely based on the non-linear behavior discussed previously. The model may need to determine which temperature regime an individual prediction falls in before producing an output. Temperature is also negatively correlated with NO_2 concentration, with higher concentrations in the colder months. As a result, running the calibration script on exclusively cold months will produce models that rely more heavily on the no2_wrk_aux variable (as non-linear behavior is less prevalent in the cold).

Using these importance values are also useful when diagnosing hardware issues with sensors, as no2_wrk_aux and temperature should always be the highest importance values. Sites where that is not the case typically indicate hardware issues with the Alphasense sensors.

Spatial Validation

Temporal validation sets have been discussed previously to test hyperparameter combinations that prevent overfitting. Some experimentation has been done with (what I call) a spatial validation set. This method is used by the Winter et al. group to test their calibration while excluding a reference station, then comparing the calibrated LCS data to the collocated reference site data.

I ran an experiment where the PHA Quant site was removed from all steps of the calibration process, including the precision and accuracy correction. The PHA BEACO$_2$N was calibrated to the next nearest Quant (PEMA), and the final data was compared to the unseen PHA Quant data. Results were successful and are depicted in figure 20.

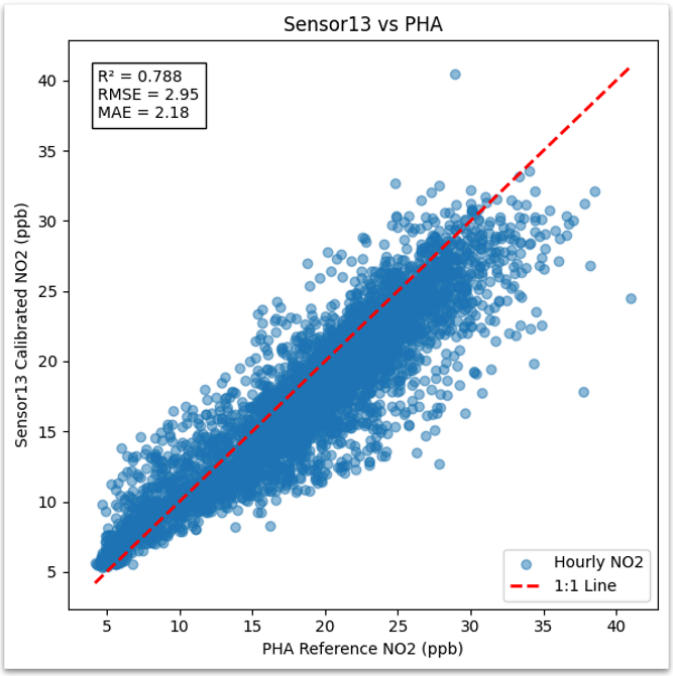


Figure 20: Calibration results when excluding PHA as a spatial validation set.

This experiment was run at a single site, but future experiments should be conducted that remove other sites from the calibration process. I would expect PEMA to perform well, while DPW may struggle a bit due to hardware issues. MJF would certainly underperform due to the shifted concentration distribution.

Next Steps

There are many ways that I would like to improve the script, some being small fixes and some being larger conceptual ideas that can be incorporated into the calibration script. Some of these may be incorporated in a final "Version 1" form of the script, while some may need to be picked up later by myself or another researcher. The following are listed in no particular order:

- Rounding issue in the table in the PDF. As seen in figure 10, there is a persistent bug that is preventing the importance values from being rounded down. This will be a bug I will try to fix soon but is only a small visual error.
- Adding interpolation for missing points for BEACO$_2$Ns and Quants. Likely a linear interpolation - I am not sure how much this will improve readings as missing data from both BEACO$_2$N and Quant tend to be for longer than just a few hours (I would not want to interpolate over periods longer than one or two hours).
- Incorporating number of hours since deployment as model parameter. Grace has included this in past calibration attempts of the Breathe Providence network to account for network drift. In the context of XGBoost I fear this may cause some overfitting and may not work well when considering time periods before the Quants were online.
- Adjusting MJF calibration. This would be reconsidering the low-RSD time period identification for sites that are calibrated to the MJF Quant. As seen in figure 14, lower concentration values are overrepresented with the traditional approach. I think for those two sites it may be more beneficial to do a simple random train-test split due to how different the underlying distribution of NO_2 is. This may also ease how stark the NO_2 gradient is across the city.
- Calibrating BEACO$_2$N data before Quants were deployed. There is still analysis to be done that shows this can be effective, but ideally a model that has been fed a wide variety of training data (particularly data that covers all seasons) can still be effective. The only direct concern would be sensor drift not being accounted for. It would also be difficult to verify the accuracy of these calibrations for almost all sites, as there would only be one true reference site online to verify calibration against at MJF DEM. Although, this site was offline from April 2024 - August 2025. It would be best to analyze the calibrated data and look for anomalies.
- Adjusting model parameters. As mentioned, co_wrk_aux was never used by the Winter et al. group and perhaps could be the result of overfitting. It would be useful to analyze model testing performance when including/excluding these values, as well as testing other parameters.
- Removing data at specific sites. The Breathe Providence OneNote document highlights when sensors were not on site. This data needs to be removed from the beginning of the calibration pipeline rather than later, as it can still impact the accuracy and precision steps of the calibration process.

- Add residual plots to the PDF report. These are visuals that are included in the Winter et al. paper and are useful to understand how bias changes as other parameters change.
- Print RSD cutoffs at some point during the calibration process. It might be useful for the user to understand what the cutoffs are before and after the 15ppb data split to understand how well-mixed the air is when calibrating, and perhaps how it changes when more or less data is used in the models.
- Incorporating Quant data before October 2025. Currently, the earliest date allowed to be entered into the calibration script is 10/28/2025 as this was most recent date that all four Quants were online continuously. However, since late 2024, three of the four Quants (DPW, PEMA, PHA) were online and collecting data. For those three sites, the precision and accuracy correction can occur and calibrated data can be produced. The only shortcoming of this is that Blackstone and MJF BEACO$_2$Ns cannot be calibrated as the MJF Quant was not online.
    - In the short term, I will likely calibrate this data separately and tie it into data produced by the script so that we will have calibrated NO_2 data on hand for all of 2025. A lot of code would need to be refactored to incorporate this change into the script.
- Data availability calendar. For each site produce a chart that shows when data is missing on a one-dimensional time series. This is useful to understand when sites were offline at certain times.
- Further spatial validation at other Quant sites. See previous section.

The best way to track which of these changes are implemented is to check pushes to the GitHub repository. If there are any specific questions, feel free to reach out to me by email at john_lardner@brown.edu.